eRAN

LTE Spectrum Coordination Feature Parameter Description

Issue 01

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Contents

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes

  Changes in functions and their corresponding parameters

- Editorial changes

  Improvements or revisions to the documentation

1.1 eRAN21.1 01 (2025-03-10)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Revised descriptions in this document.

1.2 eRAN21.1 Draft B (2025-02-10)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Added the network impacts of LTE spectrum coordination. For details, see **4.2.2 Impacts**.

1.3 eRAN21.1 Draft A (2024-12-31)

This issue introduces the following changes to eRAN20.1 03 (2024-11-29).

Technical Changes

None

Editorial Changes

None

2 About This Document

2.1 General Statements

Purpose

This document is intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Feature Differences Between RATs

The feature difference section only describes differences in switches or principles.

Unless otherwise stated, descriptions in this document apply to all RATs. If a description does not apply to all RATs, the specific RAT that it does apply to will be stated.

For example, in the statement "TDD cells are compatible with enhanced MU-MIMO", "TDD cells" indicates that this function cannot be used in non-TDD cells.

2.2 Applicable RAT

This document applies to FDD/TDD.

2.3 Features in This Document

This document describes the following FDD features.

Feature ID	Feature Name	Chapter/Section
LCOFD-131312	LTE Spectrum Coordination (LTE FDD)	• **4 LTE Spectrum Coordination** • **5 LTE Spectrum Coordination Enhancement**

This document describes the following TDD features.

Feature ID	Feature Name	Chapter/Section
TDLCOFD-131312	LTE Spectrum Coordination (LTE TDD)	• **4 LTE Spectrum Coordination** • **5 LTE Spectrum Coordination Enhancement**

2.4 Feature Differences Between FDD and TDD

FDD Feature	TDD Feature	Difference	Chapter/Section
LCOFD-131312 LTE Spectrum Coordination (LTE FDD)	TDLCOFD-131312 LTE Spectrum Coordination (LTE TDD)	None	• **4 LTE Spectrum Coordination** • **5 LTE Spectrum Coordination Enhancement**

3 Overview

In multi-band LTE networks, high frequencies usually encounter uplink usage limitation while downlink is still available. This limitation may even affect the deployment of high frequencies. LTE spectrum coordination lifts the downlink usage of carrier aggregation (CA) UEs at the uplink coverage edge of high frequencies.

4 LTE Spectrum Coordination

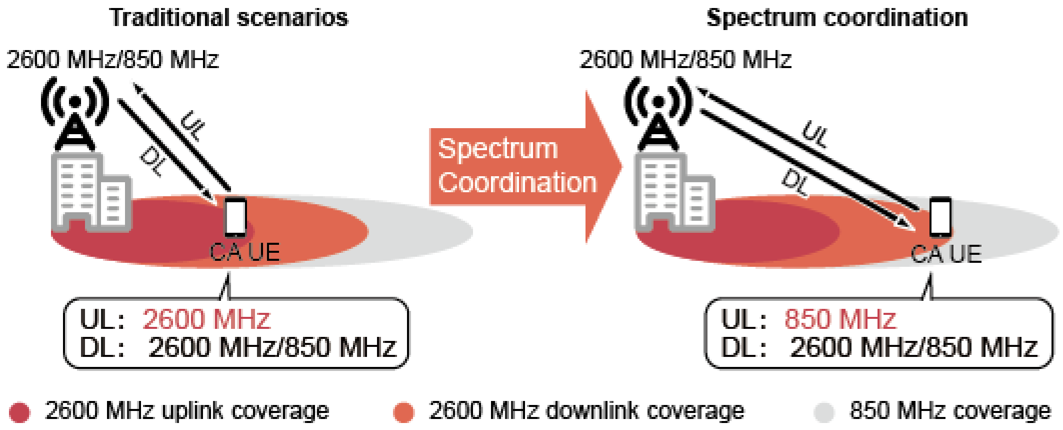
4.1 Principles

When a UE in the CA state is located at the uplink coverage edge of a high-frequency serving cell and the uplink signal to interference plus noise ratio (SINR) used in scheduling of the UE becomes unsatisfactory, the eNodeB evaluates whether any low-frequency secondary cell (SCell) of the UE can provide better uplink performance. If an SCell can provide better performance, the eNodeB hands over the UE from the primary cell (PCell) to this SCell and simultaneously configures the original PCell as an SCell for the UE.

The uplink SINR used in scheduling refers to that for initial selection of modulation and coding schemes (MCSs). For details, see *Uplink Scheduling*.

Figure 4-1 illustrates an example of LTE spectrum coordination. In this example, the CA UE was served by a PCell in the 2600 MHz band and an SCell in the 850 MHz band before LTE spectrum coordination.

Figure 4-1 Example of LTE spectrum coordination



This function is enabled when all the following conditions are met:

- The **HoAdmitSwitch** option of **CellMlbHo**.***MlbMatchOtherFeatureMode*** parameter is deselected.

- The **HoWithSccCfgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is selected.
- The **SpectrumCoordinationSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected.

This function is compatible with all CA band combinations for downlink 2CC to 8CC aggregation.

This function does not take effect for UEs that each have two component carriers (CCs) aggregated for both uplink and downlink. That is because uplink SCells will not be selected as target serving cells for LTE spectrum coordination. To be specific, for CA UEs with two CCs aggregated in the uplink and *n* CCs aggregated in the downlink:

- If *n* is equal to 2, LTE spectrum coordination does not take effect.
- If *n* is greater than 2 but does not exceed 8, LTE spectrum coordination takes effect.

After this function is enabled, the following options must be selected:

- The **CaEnhancedPreAllocSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter and the **Dl2CCAckResShareSw** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter: Selecting these options prevents the traffic volume in low-frequency cells from increasing.
- The **SccA2RmvSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter: Selecting this option prevents the possibility that UEs do not report the measurement results of SCells or the strongest intra-frequency neighboring cells of SCells.

When a UE in the CA state is transmitting uplink data in its uplink serving cell and the uplink SINR used in scheduling in that cell falls below the threshold specified by the **CaMgtCfg**.***SpectrumCoordSinrThld*** parameter, the inter-frequency handover and carrier management procedures described in this section take place.

Inter-Frequency Handover

The eNodeB evaluates the uplink channel quality and downlink signal strength in the downlink SCells of the CA UE, whether any downlink SCell meets PCell conditions, and also the CA capabilities of the UE. Based on the evaluation, the eNodeB determines whether any downlink SCell is suitable to be the target cell of a handover.

- If the downlink signal strength of a downlink SCell of the UE reaches the RSRP threshold for event A4 and is higher than all of its intra-frequency neighboring cells, this SCell is suitable to be the target cell of an inter-frequency handover, after which the target cell will take the role of PCell. If either of these conditions is not met, this SCell is not suitable to be the target cell.

  It is recommended that the value of the **CaGroupCell.*PCellA4RsrpThd*** or **PccFreqCfg.*PccA4RsrpThd*** parameter (used in CA-group-based or adaptive CA, respectively) be greater than the value of the **InterFreqHoGroup**.***InterFreqHoA4ThdRsrp*** parameter to prevent a decrease in the handover success rate.
- For details about the PCell conditions, see *Carrier Aggregation*.

If there are no downlink SCells that meet the preceding conditions, the eNodeB does not initiate an inter-frequency handover procedure. If one or more downlink SCells meet the preceding conditions, the eNodeB proceeds as follows:

- If there is only one candidate downlink SCell, the eNodeB performs a handover to that cell.
- If there are two or more candidate downlink SCells, the eNodeB selects one from them as a target cell by adhering to the following rules:
  - With intelligent selection of serving cell combinations disabled (controlled by the **CaSmartSelectionSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter)
    - The eNodeB selects the cell with the highest PCell priority (specified by the **CaGroupCell.*PreferredPCellPriority*** parameter in CA-group-based configuration mode) or PCC priority (specified by the **PccFreqCfg.*PreferredPccPriority*** parameter in adaptive configuration mode).
    - If there are two or more cells with the highest PCell or PCC priority, the eNodeB selects the cell with the largest bandwidth.
    - If there are two or more cells with the same bandwidth, the eNodeB selects the cell whose center frequency is the lowest.
  - With intelligent selection of serving cell combinations enabled
    - The eNodeB selects the cell with the highest downlink air interface capability. For how to calculate the air interface capability, see *Carrier Aggregation*.
    - If there are two or more cells with the same downlink air interface capability, the eNodeB selects the cell with the highest PCell priority (specified by the **CaGroupCell.*PreferredPCellPriority*** parameter in CA-group-based configuration mode) or PCC priority (specified by the **PccFreqCfg.*PreferredPccPriority*** parameter in adaptive configuration mode).
    - If there are two or more cells with the highest PCell or PCC priority, the eNodeB selects the cell with a larger bandwidth.
    - If there are two or more cells with the same bandwidth, the eNodeB selects the cell whose center frequency is the lowest.

After the handover, the original SCell becomes a PCell and the original PCell becomes an SCell for the UE.

When the UE is camping on the new PCell, the carrier management procedure takes place if CA is working in adaptive mode rather than in CA-group-based mode.

Carrier Management

The eNodeB takes different actions depending on the setting of the **SccFreqCfg**.***SccA2RsrpThldExtendedOfs*** parameter:

- Set to **0**

The eNodeB delivers A2 measurement configurations related to SCells to the UE. After receiving an A2 measurement report that contains an SCell, the eNodeB sends an RRC Connection Reconfiguration message to remove the SCell for the UE. For details, see *Carrier Aggregation*.

- Set to a non-zero value
  - With intelligent selection of serving cell combinations enabled
    1. The eNodeB delivers A2 measurement configurations related to SCells to the UE.

       Threshold for event A2 = **CaMgtCfg**.***CarrAggrA2ThdRsrp*** + **SccFreqCfg**.***SccA2Offset***
    2. The UE sends an A2 measurement report containing an SCell to the eNodeB.
    3. The eNodeB evaluates whether the SCell's RSRP in the A2 measurement report meets another event A2 entering condition, in which a new threshold for event A2 is used.

       New threshold for event A2 = **CaMgtCfg**.***CarrAggrA2ThdRsrp*** + **SccFreqCfg**.***SccA2Offset*** + **SccFreqCfg**.***SccA2RsrpThldExtendedOfs***

       If the RSRP meets the new condition, the eNodeB removes this SCell for the UE. The procedure ends.

       If the RSRP does not meet the new condition, the eNodeB deletes the previous A2 measurement configuration related to this SCell. Meanwhile, the eNodeB delivers the new A2 measurement configuration and an A1 measurement configuration related to this SCell to the UE.

       Threshold for event A1 = **CaMgtCfg**.***CarrAggrA4ThdRsrp*** + **SccFreqCfg**.***SccA4Offset***
    4. Before the UE sends any more A2 or A1 measurement reports related to this SCell, intelligent selection of serving cell combinations takes place to select a better SCell for the UE.

       If the UE sends an A2 measurement report containing this SCell again to the eNodeB, the eNodeB removes this SCell for the UE.

       If the UE sends an A1 measurement report containing this SCell to the eNodeB, the eNodeB releases the new A2 measurement configuration and delivers the previous A2 measurement configurations to the UE. The threshold for event A2 is changed back to the sum of **CaMgtCfg**.***CarrAggrA2ThdRsrp*** and **SccFreqCfg**.***SccA2Offset***.
  - With intelligent selection of serving cell combinations disabled
    1. The eNodeB delivers A2 measurement configurations related to SCells to the UE.

       Threshold for event A2 = **CaMgtCfg**.***CarrAggrA2ThdRsrp*** + **SccFreqCfg**.***SccA2Offset*** + **SccFreqCfg**.***SccA2RsrpThldExtendedOfs***
    2. After receiving an A2 measurement report containing an SCell from the UE, the eNodeB removes this SCell for the UE.

4.2 Network Analysis

4.2.1 Benefits

This function is suitable for CA-enabled multi-band networks in either of the following scenarios (a larger spacing between high and low frequencies brings greater benefits):

- Scenario 1: Both high- and low-frequency carriers have been deployed. The percentage of PUSCH MCSs 0 to 3 is greater than 10% in high-frequency cells, uplink interference to low-frequency cells minus that to the high-frequency cells is less than 5 dB, and the penetration rate of CA UEs exceeds 25%.
  - The average **User Downlink Average Throughput** across the network increases.
  - If the proportion of CA UEs on the network remains unchanged after this function is enabled, the overall average downlink throughput of all CA UEs on the network increases. However, if that proportion rises, the amount of the increase is reduced. The overall average downlink throughput of all CA UEs on the network even drops.

    Average downlink throughput of CA UEs = (**L.Thrp.bits.DL.CAUser** – **L.Thrp.bits.DL.LastTTI.CAUser**)/**L.Thrp.Time.DL.RmvLastTTI.CAUser**
- Scenario 2: Only a high-frequency carrier has been deployed. This function is enabled when a low-frequency carrier is being deployed.
  - The average **User Downlink Average Throughput** across the network increases.
  - The average **User Uplink Average Throughput** across the network increases.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

LTE spectrum coordination has the following impacts on the network:

- Key performance indicator (KPI) values change.

  Before handovers, the channel quality may differ between PCells and SCells of UEs. After an inter-frequency handover for LTE spectrum coordination, the channel quality indicator (CQI) changes.
  - If the original SCell provides better channel quality than the original PCell, network KPIs will improve after the handover. The KPIs include uplink and downlink cell throughput, uplink and downlink initial block error rate (IBLER), uplink and downlink average MCS indexes, radio resource control (RRC) connection reestablishment rate, and service drop rate.
  - If the original SCell provides lower channel quality than the original PCell, these KPIs will deteriorate.

  Due to changes in the PCell after a handover, the number of CA UEs and traffic related to the PCell and SCells may change, as indicated by counters **L.Traffic.User.PCell.DL.Avg**, **L.Traffic.User.SCell.DL.Avg**, **L.CA.Traffic.bits.DL.PCell**, and **L.CA.Traffic.bits.DL.SCell**.
- The number of handovers increases, which causes the number of UEs in the discontinuous reception (DRX) state to drop. For voice service UEs, the mean opinion score (MOS) probably decreases.

- After the **HoWithSccCfgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is selected, the SCell configuration success rate changes as affected by the handover success rate.

  The SCell configuration success rate can be calculated using the **L.CA.DLSCell.Add.Succ** and **L.CA.DLSCell.Add.Att** counter values.

4.3 Requirements

4.3.1 Licenses

RAT	Feature ID	Feature Name	Model	Sales Unit
FDD	LCOFD-131312	LTE Spectrum Coordination (LTE FDD)	LT1SUNPLTE00	per cell
TDD	TDLCOFD-131312	LTE Spectrum Coordination (LTE TDD)	LT1SLTESPC00	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
FDD TDD	Carrier aggregation	CA works in multiple scenarios. Its function switch varies depending on scenarios. For details, see *Carrier Aggregation*.	*Carrier Aggregation*

RAT	Function Name	Function Switch	Reference
FDD TDD	SCell configuration during handovers	**HoWithSccCfgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference
FDD TDD	Admission control of MLB triggering over unnecessary handovers	**HoAdmitSwitch** option of the **CellMlbHo**.***MlbMatchOtherFeatureMode*** parameter	*Mobility Management in Connected Mode* *Intra-RAT Mobility Load Balancing*

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
FDD	Uplink full-antenna reception	**UL_COVERAGE_BOOST_SW** option of the **SectorSplitGroup**.***SectorSplitSwitch*** parameter	*Massive MIMO Enhancements (FDD)*	After uplink full-antenna reception takes effect, the proportion of CA UEs at the uplink coverage edge of massive MIMO carriers decreases, causing a possible change in the number of inter-frequency handovers triggered for LTE spectrum coordination.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	SCC coverage threshold adaptation	**SCC_ADAPT_COV_THLD_SW** option of the **MultiCarrUnifiedSch**.***MultiCarrierUnifiedSchSw*** parameter	*Multi-carrier Unified Scheduling*	When SCC coverage threshold adaptation is enabled, carrier management in LTE spectrum coordination does not take effect.
FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **LTE_NR_SPECTRUM_SHR**	*LTE FDD and NR Dynamic Spectrum Sharing*	Fewer LTE uplink RBs are available. This affects the proportion of UEs for which LTE spectrum coordination takes effect.
FDD	LTE FDD and NR Uplink Spectrum Sharing	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **LTE_NR_UPLINK_SPECTRUM_SHR**	*LTE FDD and NR SUL Dynamic Spectrum Sharing*	Fewer LTE uplink RBs are available. This affects the proportion of UEs for which LTE spectrum coordination takes effect.
FDD	DSS and NR Flexible Refarming	**DSS_FLEX_REFARM_SW** option of the **SpectrumCloud**.***SpectrumCloudEnhSwitch*** parameter	*LTE FDD and NR Dynamic Spectrum Sharing*	After the carriers of the LTE cells engaged in spectrum sharing are shut down, fewer cells can be selected for spectrum coordination, which therefore may take effect for fewer UEs.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	NSA PCC anchoring	**NSA_PCC_ANCHORING_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter	*EPC-based NSA Performance Enhancement*	LTE spectrum coordination may cause an NSA UE to be handed over from an NSA PCC anchor to a non-anchor carrier. To prevent such handovers, you can deselect the **SPCT_COORD_INTER_FREQ_HO_SW** option of the **NsaDcMgmtConfig**.***NsaDcUeLteFunActivationSw*** parameter so that LTE spectrum coordination does not take effect for NSA UEs.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For FDD, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations)

For TDD, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Boards

LBBPc boards cannot be used for this function.

RF Modules

No requirements

4.3.4 Others

UEs must support CA.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-1 and **Table 4-2** describe the parameters used for function activation and optimization, respectively.

Table 4-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
MLB Match Other Feature Mode	**CellMlbHo.*MlbMatchOtherFeatureMode***	**HoAdmitSwitch**	Deselect this option for the original PCells and SCells of UEs.
CA Algorithm Switch	**ENodeBAlgoSwitch.*CaAlgoSwitch***	**HoWithSccCfgSwitch**	Select this option.
Cell Level CA Algorithm Switch	**CaMgtCfg.*CellCaAlgoSwitch***	**SpectrumCoordinationSwitch**	Select this option for the original PCells and SCells of UEs.
CA Algorithm Extend Switch	**ENodeBAlgoSwitch.*CaAlgoExtSwitch***	**CaEnhancedPreAllocSwitch**	Select this option.
PUCCH algorithm switch	**CellAlgoSwitch.*PucchAlgoSwitch***	**Dl2CCAckResShareSw**	Select this option.
CA Algorithm Switch	**ENodeBAlgoSwitch.*CaAlgoSwitch***	**SccA2RmvSwitch**	Select this option.

Table 4-2 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
Spectrum Coordination SINR Threshold	**CaMgtCfg.*SpectrumCoordSinrThld***	Set this parameter to its recommended value for the original PCells of UEs.

Parameter Name	Parameter ID	Setting Notes
SCC A2/A4 RSRP Threshold Extended Offset	**SccFreqCfg**.***SccA2RsrpThldExtendedOfs***	Set this parameter to its recommended value.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Enabling LTE spectrum coordination (taking downlink 3CC aggregation as an example)
MOD CELLMLBHO: LocalCellId=0, MlbMatchOtherFeatureMode=HoAdmitSwitch-0;
MOD CELLMLBHO: LocalCellId=1, MlbMatchOtherFeatureMode=HoAdmitSwitch-0;
MOD CELLMLBHO: LocalCellId=2, MlbMatchOtherFeatureMode=HoAdmitSwitch-0;
MOD CELLALGOSWITCH: LocalCellId=0, PucchAlgoSwitch=Dl2CCAckResShareSw-1;
MOD CELLALGOSWITCH: LocalCellId=1, PucchAlgoSwitch=Dl2CCAckResShareSw-1;
MOD CELLALGOSWITCH: LocalCellId=2, PucchAlgoSwitch=Dl2CCAckResShareSw-1;
MOD ENODEBALGOSWITCH: CaAlgoSwitch=HoWithSccCfgSwitch-1&SccA2RmvSwitch-1,
CaAlgoExtSwitch=CaEnhancedPreAllocSwitch-1;
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=SpectrumCoordinationSwitch-1;
MOD CAMGTCFG: LocalCellId=1, CellCaAlgoSwitch=SpectrumCoordinationSwitch-1;
MOD CAMGTCFG: LocalCellId=2, CellCaAlgoSwitch=SpectrumCoordinationSwitch-1;
```

Optimization Command Examples

```
//Setting the SINR threshold for LTE spectrum coordination (taking downlink 3CC aggregation as an
example)
MOD CAMGTCFG: LocalCellId=0, SpectrumCoordSinrThld=-2;
MOD CAMGTCFG: LocalCellId=1, SpectrumCoordSinrThld=-2;
MOD CAMGTCFG: LocalCellId=2, SpectrumCoordSinrThld=-2;

//Setting the extended offset relative to the RSRP threshold for event A2 that controls SCell removal (taking
downlink 3CC aggregation as an example)
MOD SCCFREQCFG: PccDlEarfcn=123, SccDlEarfcn=567, SccA2RsrpThldExtendedOfs=-10;
MOD SCCFREQCFG: PccDlEarfcn=123, SccDlEarfcn=890, SccA2RsrpThldExtendedOfs=-10;
MOD SCCFREQCFG: PccDlEarfcn=567, SccDlEarfcn=123, SccA2RsrpThldExtendedOfs=-10;
MOD SCCFREQCFG: PccDlEarfcn=567, SccDlEarfcn=890, SccA2RsrpThldExtendedOfs=-10;
MOD SCCFREQCFG: PccDlEarfcn=890, SccDlEarfcn=123, SccA2RsrpThldExtendedOfs=-10;
MOD SCCFREQCFG: PccDlEarfcn=890, SccDlEarfcn=567, SccA2RsrpThldExtendedOfs=-10;
```

Deactivation Command Examples

```
//Disabling LTE spectrum coordination (taking downlink 3CC aggregation as an example)
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=SpectrumCoordinationSwitch-0;
MOD CAMGTCFG: LocalCellId=1, CellCaAlgoSwitch=SpectrumCoordinationSwitch-0;
MOD CAMGTCFG: LocalCellId=2, CellCaAlgoSwitch=SpectrumCoordinationSwitch-0;
```

4.4.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.

- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  NOTE

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Observe the counters listed in **Table 4-3**. If their values increase, LTE spectrum coordination has taken effect.

Table 4-3 Counters used to verify activation of LTE spectrum coordination

Counter ID	Counter Name
1526729994	**L.HHO.InterFreq.ULquality.PrepAttOut**
1526729996	**L.HHO.InterFreq.ULquality.ExecAttOut**
1526729998	**L.HHO.InterFreq.ULquality.ExecSuccOut**
1526729995	**L.HHO.InterFddTdd.ULquality.PrepAttOut**
1526729997	**L.HHO.InterFddTdd.ULquality.ExecAttOut**
1526729999	**L.HHO.InterFddTdd.ULquality.ExecSuccOut**

4.4.3 Network Monitoring

Monitor the counters listed in **Table 4-4** to evaluate performance of LTE spectrum coordination.

Table 4-4 Counters used to monitor performance of LTE spectrum coordination

Counter ID	Counter Name
1526728426	**L.Traffic.User.PCell.DL.Avg**
1526728427	**L.Traffic.User.SCell.DL.Avg**
1526732658	**L.CA.Traffic.bits.DL.PCell**
1526729259	**L.CA.Traffic.bits.DL.SCell**
1526728564	**L.Thrp.bits.DL.CAUser**
1526740440	**L.Thrp.bits.DL.LastTTI.CAUser**

Counter ID	Counter Name
1526740441	**L.Thrp.Time.DL.RmvLastTTI.CAUser**
1526729045	**L.CA.DLSCell.Add.Att**
1526729046	**L.CA.DLSCell.Add.Succ**
1526746999	**L.Thrp.DL.BitRate.Samp.CaUe.Index0**
1526747000	**L.Thrp.DL.BitRate.Samp.CaUe.Index1**
1526747001	**L.Thrp.DL.BitRate.Samp.CaUe.Index2**
1526747002	**L.Thrp.DL.BitRate.Samp.CaUe.Index3**
1526747003	**L.Thrp.DL.BitRate.Samp.CaUe.Index4**
1526747004	**L.Thrp.DL.BitRate.Samp.CaUe.Index5**
1526747005	**L.Thrp.DL.BitRate.Samp.CaUe.Index6**
1526747006	**L.Thrp.DL.BitRate.Samp.CaUe.Index7**
1526747007	**L.Thrp.DL.BitRate.Samp.CaUe.Index8**
1526747008	**L.Thrp.DL.BitRate.Samp.CaUe.Index9**

5 LTE Spectrum Coordination Enhancement

5.1 Principles

5.1.1 LTE Spectrum Coordination Enhancement for WBB UEs

In addition to LTE spectrum coordination described in **4 LTE Spectrum Coordination**, LTE spectrum coordination enhancement applies to wireless broadband (WBB) UEs.

When a CA-capable WBB UE is located at the coverage edge of its high-frequency serving cell and has low downlink RSRP in this cell, the eNodeB evaluates whether there is any low-frequency cell that can provide better performance. If there is, the eNodeB hands over the UE from its PCell to that low-frequency cell and simultaneously configures the original PCell as an SCell for the UE.

NOTE

There are two methods for eNodeBs to identify WBB UEs: based on subscriber profile IDs (SPIDs) and based on QoS class identifiers (QCIs). For details, see *WBB*.

LTE spectrum coordination enhancement for WBB UEs is enabled when all of the following conditions are met:

- Low-frequency cells that do not allow access of WBB UEs have been defined as MBB service cells (by setting the **Cell**.***SpecifiedCellFlag*** parameter to **MBBSERCELL**).
- The **HoWithSccCfgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is selected.
- The **WbbCaMultiCarrierCoordSw** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected.

This function takes effect only for WBB CA UEs that have entered non-MBB service cells by initial access.

When data transmission is ongoing for a WBB CA UE in its PCell and the downlink RSRP in the PCell is lower than a certain threshold, the following takes place:

1. The eNodeB delivers A2 measurement configurations related to the PCell to the UE.

   The A2 threshold varies depending on the type of the WBB UE, as listed in **Table 5-1**.

   Table 5-1 A2 thresholds for different types of WBB UEs

Type of WBB UE	A2 Threshold
WBB UEs identified based on SPIDs	**InterFreqHoGroup**.***InterFreqHoA2ThdRsrp*** + **SpidCfg**.***InterFreqHoA2RsrpThdFactor*** + **CellWttxParaCfg**.***WbbMultiCarrierCoordA2Ofs***
WBB UEs identified based on QCIs	**InterFreqHoGroup**.***InterFreqHoA2ThdRsrp*** + **CellWttxParaCfg**.***WbbMultiCarrierCoordA2Ofs***

   If both the conditions for event A2 triggered by this function and the conditions for event A2 related to coverage-based handovers are met, event A2 related to coverage-based handovers takes precedence. As a result, this function does not take effect in this situation.

2. The UE sends A2 measurement reports to the eNodeB.

   A handover is triggered to change the PCell if all of the following conditions are met:

   – The UE is not running an emergency call.
   – The UE is not running an evolved multimedia broadcast/multicast service (eMBMS).
   – The UE does not have a QCI 1, 65, or 66 bearer.

   If any of these conditions is not met, the procedure ends.

3. The eNodeB delivers A5 measurement configurations to the UE based on the value of the **IntraRatHoComm**.***InterFreqHoA4TrigQuan*** parameter. The measurement objects are preferentially non-MBB frequencies.

   The triggering quantity parameter has the following values:

   – **RSRP**: The A5 measurement configurations include only RSRP configurations. Threshold 1 for event A5 is fixed at –43 dBm.
   – **RSRQ**: The A5 measurement configurations include only RSRQ configurations. Threshold 1 for event A5 is fixed at –3 dB.
   – **BOTH**: The A5 measurement configurations include both RSRP and RSRQ configurations.

   Threshold 2 for event A5 varies depending on the type of WBB UE, as listed in **Table 5-2** and **Table 5-3**.

Table 5-2 Threshold 2 for event A5 for WBB UEs identified based on SPIDs

SCell Configuration Mode	Triggering Quantity for Event A5	Threshold 2 for Event A5
CA-group-based SCell configuration	RSRP	Max{**CaGroupCell**.***PCellA4RsrpThd***, **InterFreqHoGroup**.***InterFreqHoA2ThdRsrp*** + **SpidCfg**.***InterFreqHoA2RsrpThdFactor*** + **CellWttxParaCfg**.***WbbMultiCarrier-CoordA2Ofs*** + 2}
	RSRQ	Max{**CaGroupCell**.***PCellA4RsrqThd***, **InterFreqHoGroup**.***InterFreqHoA2ThdRsrq*** + **SpidCfg**.***InterFreqHoA2RsrqThdFactor*** + **CellWttxParaCfg**.***WbbMultiCarrier-CoordA2Ofs*** + 2}
Adaptive SCell configuration	RSRP	Max{**PccFreqCfg**.***PccA4RsrpThd***, **InterFreqHoGroup**.***InterFreqHoA2ThdRsrp*** + **SpidCfg**.***InterFreqHoA2RsrpThdFactor*** + **CellWttxParaCfg**.***WbbMultiCarrier-CoordA2Ofs*** + 2}
	RSRQ	Max{**PccFreqCfg**.***PccA4RsrqThd***, **InterFreqHoGroup**.***InterFreqHoA2ThdRsrq*** + **SpidCfg**.***InterFreqHoA2RsrqThdFactor*** + **CellWttxParaCfg**.***WbbMultiCarrier-CoordA2Ofs*** + 2}

Table 5-3 Threshold 2 for event A5 for WBB UEs identified based on QCIs

SCell Configuration Mode	Triggering Quantity for Event A5	Threshold 2 for Event A5
CA-group-based SCell configuration	RSRP	Max{**CaGroupCell**.***PCellA4RsrpThd***, **InterFreqHoGroup**.***InterFreqHoA2ThdRsrp*** + **CellWttxParaCfg**.***WbbMultiCarrier-CoordA2Ofs*** + 2}

SCell Configuration Mode	Triggering Quantity for Event A5	Threshold 2 for Event A5
	RSRQ	Max{**CaGroupCell**.***PCellA4RsrqThd***, **InterFreqHoGroup**.***InterFreqHoA2ThdRsrq*** + **CellWttxParaCfg**.***WbbMultiCarrier-CoordA2Ofs*** + 2}
Adaptive SCell configuration	RSRP	Max{**PccFreqCfg**.***PccA4RsrpThd***, **InterFreqHoGroup**.***InterFreqHoA2ThdRsrp*** + **CellWttxParaCfg**.***WbbMultiCarrier-CoordA2Ofs*** + 2}
	RSRQ	Max{**PccFreqCfg**.***PccA4RsrqThd***, **InterFreqHoGroup**.***InterFreqHoA2ThdRsrq*** + **CellWttxParaCfg**.***WbbMultiCarrier-CoordA2Ofs*** + 2}

4. The eNodeB receives A5 measurement reports from the UE.
   - If the target cell is not an MBB service cell, the eNodeB acts as described in *WBB*.
   - If the target cell is an MBB service cell, the eNodeB checks whether this cell can be aggregated with any of the current serving cells of the UE for CA.
     - If it can, the eNodeB initiates an inter-frequency handover procedure.
     - If it cannot, the eNodeB does not initiate an inter-frequency handover procedure. The procedure ends.
5. After the UE is handed over to the MBB service cell:
   - The original PCell becomes an SCell of the UE.
   - WBB-dedicated cells can be configured as SCells.
   - MBB service cells cannot be configured as SCells for the UE.
   - The eNodeB will not remove SCells based on event A2.

   After the handover, the MBB service cell acts as the PCell for the UE, and downlink data for the UE is scheduled only in its SCells whenever possible. If all SCells fail to be configured for the UE, the UE is allowed to use the resources of the MBB service cell.

5.1.2 LTE Spectrum Coordination Enhancement for RRNs

For relay remote nodes (RRNs), LTE spectrum coordination enhancement has the following treatment on their CA in out-of-band relay scenarios:

- If the PCell of an RRN is not an MBB service cell, MBB service cells cannot be configured as SCells.
- If the PCell of an RRN is an MBB service cell, non-MBB service cells can be configured as SCells. Downlink data is scheduled only in SCells, whenever

possible. If all SCells fail to be configured for the RRN, the RRN is allowed to use the resources of the MBB service cell.

LTE spectrum coordination enhancement for RRNs is enabled when both of the following conditions are met:

- Low-frequency cells that do not allow access of WBB UEs have been defined as MBB service cells (by setting the **Cell**.***SpecifiedCellFlag*** parameter to **MBBSERCELL**).
- The **WbbCaMultiCarrierCoordSw** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected.

5.2 Network Analysis

5.2.1 Benefits

This function is recommended for networks deployed with both high-frequency WBB carriers and low-frequency MBB carriers.

Assume that the low-frequency cells are normal cells before this function is enabled on a network. Then, activation of this function increases the perceived data rate of MBB UEs.

Perceived data rate of MBB UEs = [(**L.Thrp.bits.DL** – **L.Thrp.bits.DL.LastTTI**) – (**L.Thrp.bits.DL.WBB** – **L.Thrp.bits.DL.LastTTI.WBB**)]/(**L.Thrp.Time.DL.RmvLastTTI** – **L.Thrp.Time.DL.RmvLastTTI.WBB**)

These counters are described in **Table 5-4**.

Table 5-4 Counters related to the perceived data rate of MBB UEs

Counter ID	Counter Name
1526728261	**L.Thrp.bits.DL**
1526729005	**L.Thrp.bits.DL.LastTTI**
1526745879	**L.Thrp.bits.DL.WBB**
1526745880	**L.Thrp.bits.DL.LastTTI.WBB**
1526729015	**L.Thrp.Time.DL.RmvLastTTI**
1526745881	**L.Thrp.Time.DL.RmvLastTTI.WBB**

5.2.2 Impacts

The impacts between this function and other functions are described in **5.3.2.3 Function Impacts**.

Assume that the low-frequency cells are normal cells before this function is enabled on a network. Then, activation of this function has the following impacts on the network:

- The overall average perceived data rate of CA UEs decreases.

  Perceived data rate of CA UEs = (**L.Thrp.bits.DL.CAUser** – **L.Thrp.bits.DL.LastTTI.CAUser**)/**L.Thrp.Time.DL.RmvLastTTI.CAUser**

  These counters are described in **Table 5-5**.

Table 5-5 Counters related to the perceived data rate of CA UEs

Counter ID	Counter Name
1526728564	**L.Thrp.bits.DL.CAUser**
1526740440	**L.Thrp.bits.DL.LastTTI.CAUser**
1526740441	**L.Thrp.Time.DL.RmvLastTTI.CAUser**

- CA takes effect for a smaller proportion of WBB UEs. Therefore, the value of the **L.Thrp.Time.DL.RmvLastTTI.WBB** counter, with an ID of 1526745881, probably increases.
- The perceived data rate of WBB UEs decreases.

  Perceived data rate of WBB UEs = (**L.Thrp.bits.DL.WBB** – **L.Thrp.bits.DL.LastTTI.WBB**)/**L.Thrp.Time.DL.RmvLastTTI.WBB**

  These counters are described in **Table 5-6**.

Table 5-6 Counters related to the perceived data rate of WBB UEs

Counter ID	Counter Name
1526745879	**L.Thrp.bits.DL.WBB**
1526745880	**L.Thrp.bits.DL.LastTTI.WBB**
1526745881	**L.Thrp.Time.DL.RmvLastTTI.WBB**

- The perceived data rate of RRNs decreases.

  Perceived data rate of RRNs = **L.Thrp.Relay.bits.DL**/**L.Thrp.Relay.Time.DL**

  These counters are described in **Table 5-7**.

Table 5-7 Counters related to the perceived data rate of RRNs

Counter ID	Counter Name
1526735542	**L.Thrp.Relay.bits.DL**
1526735543	**L.Thrp.Relay.Time.DL**

- The value of the **L.ChMeas.PRB.DL.PDSCH.WBBUsed.Avg** counter, with an ID of 1526745954, decreases in the low-frequency cells but increases in the high-frequency cells.
- The values of the counters listed in **Table 5-8** decrease for the low-frequency cells.

Table 5-8 Counters related to CA

Counter ID	Counter Name
1526728426	**L.Traffic.User.PCell.DL.Avg**
1526728427	**L.Traffic.User.SCell.DL.Avg**
1526732658	**L.CA.Traffic.bits.DL.PCell**
1526729259	**L.CA.Traffic.bits.DL.SCell**
1526728564	**L.Thrp.bits.DL.CAUser**
1526729045	**L.CA.DLSCell.Add.Att**
1526729046	**L.CA.DLSCell.Add.Succ**

- The number of handovers increases, and the service drop rate probably rises.
- After the **HoWithSccCfgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is selected, the SCell configuration success rate decreases as affected by the handover success rate.

  The SCell configuration success rate can be calculated using the **L.CA.DLSCell.Add.Succ** and **L.CA.DLSCell.Add.Att** counter values.

5.3 Requirements

5.3.1 Licenses

RAT	Feature ID	Feature Name	Model	Sales Unit
FDD	LCOFD-131312	LTE Spectrum Coordination (LTE FDD)	LT1SUNPLTE00	per cell
TDD	TDLCOFD-131312	LTE Spectrum Coordination (LTE TDD)	LT1SLTESPC00	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Carrier aggregation	CA works in multiple scenarios. Its function switch varies depending on scenarios. For details, see *Carrier Aggregation*.	*Carrier Aggregation*	None
FDD TDD	Specified service carrier	**Cell**.***SpecifiedCellFlag***	*WBB*	Specified service carrier is required for LTE spectrum coordination enhancement for WBB UEs.
FDD TDD	SCell configuration during handovers	**HoWithSccCfgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	SCell configuration during handovers is required for LTE spectrum coordination enhancement for WBB UEs.
FDD TDD	Carrier aggregation for out-of-band relay	Carrier aggregation for out-of-band relay works in multiple scenarios. Its function switch varies depending on scenarios. For details, see *Relay*.	*Relay*	Carrier aggregation for out-of-band relay is required for LTE spectrum coordination enhancement for RRNs.

5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference
FDD TDD	Carrier-level rate control for CA UEs	**CaUserLimitOptSwitch** option of the **CellAlgoSwitch**.***DacqEnhancementSwitch*** parameter	*Rate Control Based on User Types*

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Specified service carrier	**Cell**.***SpecifiedCellFlag***	*WBB*	After LTE spectrum coordination enhancement is enabled, WBB CA UEs can be handed over to MBB service cells. After the handovers, WBB and MBB carriers can be aggregated for CA.
FDD TDD	NSA PCC anchoring	**NSA_PCC_ANCHORING_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter	*EPC-based NSA Performance Enhancement*	NSA PCC anchoring takes precedence over LTE spectrum coordination enhancement. When both of these functions are enabled, only NSA PCC anchoring takes effect.

RAT	Function Name	Function Switch	Reference	Description
TDD	Inter-CC load transfer triggered by cell load	**DlCaLbAlgoSwitch** option of the **ENodeBAlgoSwitch**.***CaLbAlgoSwitch*** parameter	*Carrier Aggregation*	Inter-CC load transfer triggered by cell load does not take effect for WBB CA UEs or RRNs on which LTE spectrum coordination enhancement has taken effect.
TDD	Inter-CC load transfer triggered by cell load differences	**CaLoadBalancePreAllocSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	Inter-CC load transfer triggered by cell load differences does not take effect for WBB CA UEs or RRNs on which LTE spectrum coordination enhancement has taken effect.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Boards

(TDD) Cells on LBBPc boards cannot be used for this function.

(FDD) No requirements

RF Modules

No requirements

5.3.4 Others

UEs must support CA.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-9 describes the parameters used for function activation.

Table 5-9 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	**HoWithSccCfgSwitch**	Select this option.
Specified Cell Flag	**Cell**.***SpecifiedCellFlag***	None	Set this parameter to **MBBSERCELL** for low-frequency cells.
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	**WbbCaMultiCarrierCoordSw**	Select this option for low- and high-frequency cells.

Table 5-10 describes the parameters used for optimization of LTE spectrum coordination enhancement for WBB UEs.

Table 5-10 Parameters used for optimization of LTE spectrum coordination enhancement for WBB UEs

Parameter Name	Parameter ID	Setting Notes
WBB Multi-Carrier Coordination A2 Offset	**CellWttxParaCfg**.***WbbMultiCarrierCoordA2Ofs***	Set this parameter to its recommended value for low-frequency cells.
SCC A2/A4 RSRP Threshold Extended Offset	**SccFreqCfg**.***SccA2RsrpThldExtendedOfs***	Set this parameter to its recommended value.

5.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Activating LTE spectrum coordination enhancement
MOD ENODEBALGOSWITCH: CaAlgoSwitch=HoWithSccCfgSwitch-1;
MOD CELL: LocalCellId=0, SpecifiedCellFlag=MBBSERCELL;
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=WbbCaMultiCarrierCoordSw-1;
MOD CAMGTCFG: LocalCellId=1, CellCaAlgoSwitch=WbbCaMultiCarrierCoordSw-1;
```

Optimization Command Examples

```
//Setting WbbMultiCarrierCoordA2Ofs
MOD CELLWTTXPARACFG: LocalCellId=0, WbbMultiCarrierCoordA2Ofs=2;
MOD CELLWTTXPARACFG: LocalCellId=1, WbbMultiCarrierCoordA2Ofs=2;

//Setting SccA2RsrpThldExtendedOfs (The following command takes a low-frequency carrier whose
downlink EARFCN is 123 as the PCC and takes a high-frequency WBB carrier whose downlink EARFCN is
456 as an SCC.)
MOD SCCFREQCFG: PccDlEarfcn=123, SccDlEarfcn=456, SccA2RsrpThldExtendedOfs=-10;
```

Deactivation Command Examples

```
//Deactivating LTE spectrum coordination enhancement
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=WbbCaMultiCarrierCoordSw-0;
MOD CAMGTCFG: LocalCellId=1, CellCaAlgoSwitch=WbbCaMultiCarrierCoordSw-0;
```

5.4.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.
- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  NOTE

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Observe the **L.Traffic.SpecSerUser.Avg** counter, whose ID is 1526740478. If the value of this counter for the low-frequency cells decreases to a non-zero value, LTE spectrum coordination enhancement for WBB UEs has taken effect.

Observe the **L.Thrp.Relay.Time.DL** counter, whose ID is 1526735543. If the value of this counter for the low-frequency cells increases, LTE spectrum coordination enhancement for RRNs has taken effect.

5.4.3 Network Monitoring

Observe the values of counters described in **5.2 Network Analysis** after this function is enabled.

6 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, LOFD-001016 or TDLOFD-001016.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, LOFD-001016 or TDLOFD-001016.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- *Uplink Scheduling*
- *Carrier Aggregation*
- *Mobility Management in Connected Mode*
- *Intra-RAT Mobility Load Balancing*
- *WBB*
- *Rate Control Based on User Types*
- *Relay*
- *EPC-based NSA Performance Enhancement*
- *Multi-carrier Unified Scheduling*
- *Massive MIMO Enhancements (FDD)*

